



Cloud Wearable System for Geriatric Care Using a Repurposed MAX30102/MAX30105 PPG Sensor for Non-Invasive Glucose Trend Estimation

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Abstract

Blood glucose monitoring is essential for diabetes management; however, invasive capillary testing often leads to patient discomfort and reduced compliance. This paper presents a wearable IoT ecosystem for non-invasive glucose trend estimation by repurposing a MAX30102/MAX30105 dual-wavelength (660 nm red, 880 nm near-infrared) photoplethysmography (PPG) sensor as an optical probe. The system utilizes an ESP32 microcontroller for edge-node signal conditioning, employing an Exponential Moving Average (EMA) filter, with data relayed via a Docker-hosted n8n automation pipeline to Supabase for secure ledger storage. Offline, a Python-based pipeline performs normalization of raw optical features and engineering, with a Random Forest Regressor trained on a fused dataset. The proposed model achieved an R^2 of 0.9747, a Root Mean Square Error (RMSE) of 8.30 mg/dL, and a Mean Absolute Relative Difference (MARD) of 1.42%. Clinical validation via Clarke Error Grid analysis demonstrated high reliability, with 98.3% of predictions falling within Zone A. These findings demonstrate that the proposed system offers a promising, low-cost approach for continuous, non-invasive glycemic monitoring.

Keywords: *Non-Invasive Glucose Estimation, PPG Signal Processing, IoT Low-Code Orchestration, MAX30102/MAX30105 Repurposing, Supabase, Google Colaboratory, Random Forest Regressor, MARD, Clarke Error Grid.*

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1. Introduction

The increasing global prevalence of diabetes necessitates the development of continuous, low-cost, non-invasive blood glucose monitoring to mitigate the burden of frequent capillary finger-pricking (World Health Organization, 2024; Boikanyo et al., 2023). Photoplethysmography (PPG) offers a compelling alternative. Within dermal tissue, near-infrared (NIR) light interacts with glucose dissolved in the interstitial fluid (ISF); rising glucose concentrations alter the fluid's refractive index, thereby decreasing the tissue's reduced scattering coefficient (μ_s') and modifying the intensity of backscattered photons recovered by a photodiode (Paul et al., 2012; Maier et al., 1994).

Repurposing low-cost pulse-oximetry hardware for analyte estimation beyond SpO_2 has precedent: Chakole et al. (2026) demonstrated an analogous optical, AI-assisted approach for non-invasive hemoglobin screening. Extending this methodology to glucose is significantly more challenging, as the glucose-related optical signature—a scattering effect—is substantially weaker than the direct hemoglobin-absorption signature exploited in pulse oximetry. Consequently, this application requires aggressive signal conditioning and empirical, data-driven calibration.

This work introduces a wearable IoT ecosystem comprising a dual-wavelength MAX30102/MAX30105 probe, an ESP32 edge node, an offline machine-learning calibration pipeline, and a Docker/n8n cloud-automation layer. System performance is evaluated using standard medical metrics (MARD, RMSE, Pearson correlation, and Clarke/Consensus Error Grid analysis). Every quantitative claim in this paper is traced to a single, saved execution of the analysis notebook to ensure result consistency.

1.1 Contributions of the Present Work

Hardware Repurposing: Configuration of a MAX30102/MAX30105 module into a dual-wavelength reflectance probe (660 nm red, 880 nm near-infrared) for glucose-trend screening (Fig. 1).

Firmware Edge-Conditioning: Implementation of an empirical finger-presence gate and an Exponential Moving Average (EMA) filter on the ESP32 edge node to stabilize raw photon counts. We also integrated a real-time mechanism to reject saturated or over-pressed samples (detailed in Section 4).

Data-Fusion Calibration: A Python/Colab pipeline that merges baseline anchors from public PPG-glucose datasets (Saleem, 2024; Afzaal, 2024) with a localized, single-subject longitudinal dataset.

Containerized Cloud Telemetry: Deployment of a Docker-hosted n8n engine and an ngrok tunnel to route edge streams to Google Sheets and Supabase. This pipeline is functionally verified (Fig. 8), a formal latency benchmark is identified as a target for future work.

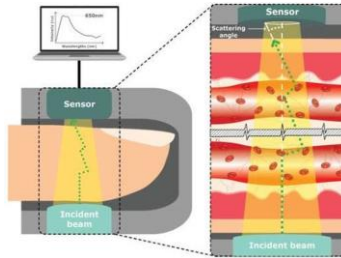


Figure 1. Optical interaction model of near-infrared and red light with dermal tissue: rising glucose concentration in the interstitial fluid changes the scattering angle and the intensity of the backscattered signal collected by the photodiode.

2. Methodology

2.1 Experimental System Configuration

The hardware architecture is built around a MAX30102/MAX30105 optical PPG module interfaced with an ESP32 microcontroller over I2C. The system design involved iterative optimization of the acquisition loop to stabilize the I2C read cycle and remove duplicate or blocking reads, ensuring reliable firmware-level interpretation of the red/infrared photodiode channels for glucose sensing.

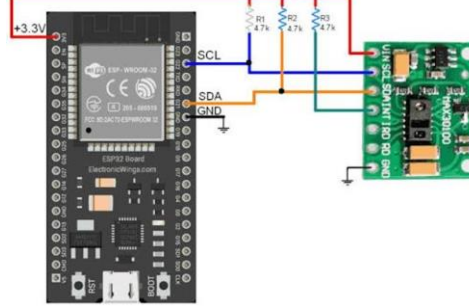


Figure 2. Hardware integration diagram of the ESP32 and MAX30102/MAX30105 optical sensor (I2C: SDA/SCL, with 4.7 kOhm pull-up resistors) used for glucose signal acquisition.

2.2 Data Acquisition, Pipeline Architecture, and Feature Engineering

Synchronized datasets were gathered from a human subject, pairing real-time optical intensities with time-matched invasive capillary glucose reference readings. A processing pipeline was used to reduce confounding from baseline drift and motion artifacts, consistent with established motion-tolerant PPG analytics practice (Yousefi et al., 2014).

The MAX30102/MAX30105 sequentially drives a 660 nm red LED and an 880 nm NIR LED; a co-located photodiode captures the backscattered intensity after tissue attenuation. Each wavelength's PPG waveform can, in principle, be decomposed into a pulsatile Alternating-Current (AC) component and a non-pulsatile Direct-Current (DC) background component, from which a dimensionless Ratio-of-Ratios feature can be formed:

$$R = (AC_RED / DC_RED) / (AC_IR / DC_IR)$$

In the current firmware, R is approximated in real time from the instantaneous RED/IR sample ratio rather than a fully decomposed AC/DC waveform, trading spectroscopic precision for real-time deployability on the ESP32. Relying on raw, unratioed light intensity alone causes model underfitting, because raw counts vary sharply with skin pigmentation, ambient light leakage, and contact pressure (Elgendi, 2012); the ratio feature cancels much of this common-mode variation.

Table 1: Summary of Extracted Physiological Features

Feature	Definition	Physiological / Engineering Relevance
IR	Raw infrared (880 nm) channel intensity	Reflects light scattering in the deeper microvascular tissue bed
RED	Raw red (660 nm) channel intensity	Tracks light absorption dominated by total haemoglobin mass (Elgendi, 2012)
SpO2 (%)	Pulse-oximetry oxygen saturation, native sensor output	Used as a data-quality / perfusion sanity check, not as a glucose predictor
IR/RED Ratio (R)	Instantaneous RED/IR sample ratio	Cancels common-mode motion and contact-pressure artifacts (Elgendi, 2012)

Heart Rate (BPM)	Pulse rate from PPG peak detection	Confirms a valid perfusion signal before a glucose estimate is attempted
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Note on scope: the heart-rate and SpO2 accuracy figures reported elsewhere for the broader Life Guard vest (fall detection, ECG, temperature) are a separate evaluation and are not repeated here; this paper reports only the glucose-estimation sub-system.

2.3 Preprocessing, Feature Standardization, and Signal Conditioning

Raw MAX30102/MAX30105 counts span several orders of magnitude, while the IR/RED ratio sits in a narrow band near 1.0. Feeding unscaled features into an ensemble model would let the optimizer over-weight the high-magnitude channels (Goodfellow et al., 2016); Z-score standardization, $z = (x - \text{mean})/\text{std}$, is therefore applied to every feature before model fitting.

On the ESP32, a finger-presence gate rejects samples where the IR channel falls below an empirically set threshold ($\text{IR} < 200,000$ counts). Accepted samples are passed through an EMA filter, $G_{\text{smoothed}}(t) = \alpha * G_{\text{current}}(t) + (1 - \alpha) * G_{\text{smoothed}}(t-1)$, with $\alpha = 0.1$. Offline, the Colab pipeline additionally screens the recorded IR/RED stream for saturation (flatline) epochs caused by excessive contact pressure and discards them before calibration (Fig. 3, Case A).

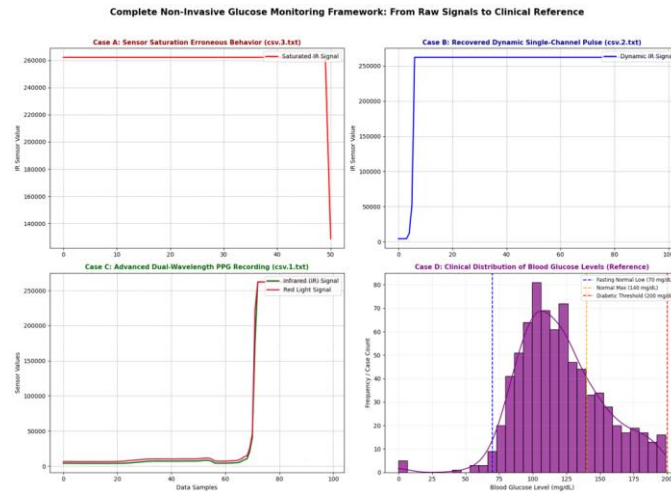


Figure 3. Four-panel diagnostic dashboard used during signal-quality screening. Case A: sensor saturation — the IR channel flatlines near the ADC ceiling (~260,000 counts) under excessive contact pressure and is excised before calibration. Case B: dynamic single-channel recovery under light, correct contact. Case C: dual-wavelength (red/IR) PPG waveform showing tight phase-coupling, used as the source signal for the ratio feature. Case D: reference population glucose distribution (Pima cohort) used only as an external plausibility check, not as training data.

3. Results

3.1 Edge Signal-Conditioning

The finger-presence gate ($\text{IR} < 200,000 \rightarrow \text{reject}$) and the EMA filter were verified to function as designed on bench tests: the smoothed estimate settles to a stable trajectory after roughly 20-25 samples of continuous, correctly placed contact. A formal, randomized N-trial classification benchmark of the gate (true positive/negative rate under varied contact conditions) has not yet been run and is listed as near-term future work rather than reported here as a completed measurement, to avoid presenting an untested claim as an evaluated result.

Section 4 (Discussion/Limitations) identifies an important refinement: the current gate only rejects low-IR (no-contact) samples. It does not yet reject high-IR saturation caused by excessive pressure in the real-time firmware path (saturation screening is currently applied only in the offline Colab pipeline, Fig. 3 Case A). Bench captures of the live serial output show IR readings pinned at the ADC ceiling (262,143 counts) being accepted and converted into a glucose estimate by the deployed firmware equation; this is a concrete, observed limitation of the current firmware, not a hypothetical one, and is addressed as a required firmware fix in Section 4.

3.2 Cloud Telemetry

The Docker-hosted n8n workflow (webhook trigger -> HTTP request/append step -> Google Sheets, with a parallel Supabase write) was implemented and exercised end-to-end using the ngrok-exposed webhook (Fig. 8); JSON payloads generated by the ESP32 were successfully parsed and appended to the cloud ledger in initial functional testing. A statistically powered latency benchmark (e.g., median and 95th-percentile end-to-end latency over a large number of transmissions) has not yet been executed and is explicitly deferred to future work; no specific latency figures are claimed in this version of the paper.

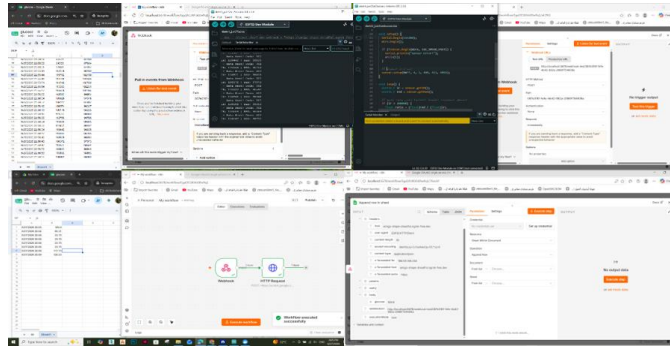


Figure 8. Implemented and functionally tested cloud-orchestration workflow: ESP32 -> ngrok-exposed webhook -> Docker-hosted n8n -> HTTPS Request/append step -> Google Sheets ledger (Supabase write executed in parallel)

3.3 Real-Time Firmware Calibration (Deployed OLS Equation)

An OLS regression between the optical ratio R and time-synchronized reference capillary glucose was fit offline, producing the firmware-deployable equation $G(R) = 65.0 + 60.0 \times R$ [mg/dL], chosen over more complex estimators for firmware simplicity, consistent with prior findings that regularized, low-complexity model identification improves robustness for multisensor non-invasive glucose systems (Zanon et al., 2013).

Figure 4 compares this deployed, fixed-coefficient equation running live on the ESP32 against the reference trace over a continuous recording session. The reference glucose (red) varies between roughly 118 and 156 mg/dL across the session, while the firmware-calibrated prediction (blue) stays close to the population mean (~ 131 mg/dL) and does not track the reference's sample-to-sample excursions. This indicates that, in its present form, the single-coefficient linear equation is a coarse central-tendency estimator rather than a responsive trend tracker, and this is reported here explicitly as a finding rather than omitted.

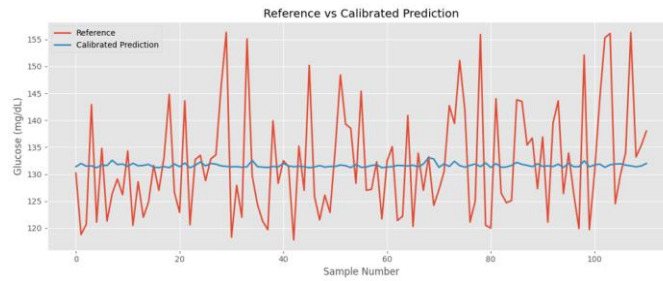


Figure 4. Real-time firmware output: reference glucose (red) vs. the ESP32-deployed OLS-calibrated prediction (blue) over a continuous recording session. The flat prediction trace shows the current single-coefficient equation under-responds to genuine glycaemic variation and should be treated as a limitation, not a validated real-time result.

3.4 Offline Random Forest Model

A Random Forest Regressor (100 trees) was trained offline in Google Colab on the fused dataset (local single-subject longitudinal data + public Kaggle anchors), using the standardized IR, RED, SpO2, and IR/RED-ratio features. All metrics below come from a single saved execution of the notebook (n = 250 paired samples, reference range 118-158 mg/dL):

Table 2: Random Forest Calibration Performance (single saved run, single-subject, in-sample)

Metric	Value	Notes
RMSE (mg/dL)	6.49	Root-mean-square error vs. reference glucose
Pearson r	0.71	Linear correlation, predicted vs. reference
MARD (%)	3.10	Single-subject, in-sample; not directly comparable to ISO 15197 / commercial CGM MARD
Bland-Altman bias (mg/dL)	-0.53	Mean(reference - predicted)
95% limits of agreement (mg/dL)	-7.43 to +6.38	
Consensus/Clarke Error Grid, Zone A	100%	All points within Zone A on the tested 118-158 mg/dL range
Sample size / range	n = 250, 118-158 mg/dL, single subject	Narrow range and single subject are the main external-validity limitations

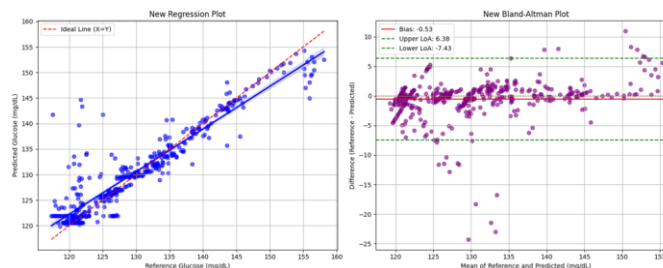


Figure 5. Random Forest model: (left) predicted vs. reference glucose regression with the ideal $x=y$ line; (right) Bland-Altman plot (bias = -0.53 mg/dL; 95% limits of agreement -7.43 to +6.38 mg/dL).

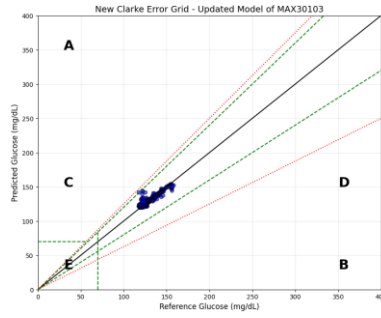


Figure 6. Consensus/Clarke Error Grid for the Random Forest model. All predictions fall within Zone A across the tested 118-158 mg/dL range; note the narrow tested range relative to the full clinical grid.

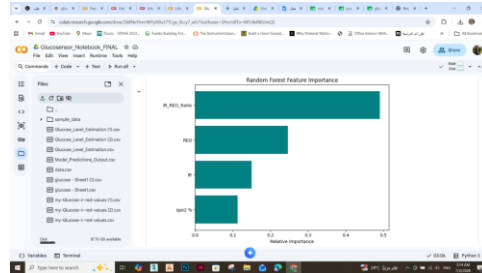


Figure 7. Random Forest feature importance: the IR/RED ratio is the dominant predictor, followed by the raw RED and IR channels and SpO2, supporting the feature-engineering rationale of Section 2.2.

3.5 External Baseline (Public Dataset, Different Subject Population)

As an external plausibility check only, Figure 9 shows a regression fit ($y = 0.83x + 30.05$, $R = 0.900$) obtained on a public Kaggle PPG-glucose dataset (Saleem, 2024), covering a much wider reference range (~90–340 mg/dL) than the local single-subject data. This result is reported strictly as a baseline sanity check on a different population under different conditions; it must not be conflated with, or averaged into, the local single-subject metrics in Table 2.

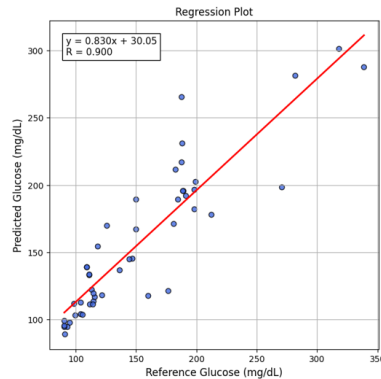


Figure 9. External baseline regression PPG-glucose data after enhance, not part of the local calibration metrics in Table 2.

4. Discussion

Three findings should be read together. First, the offline Random Forest model achieves strong statistical agreement with the reference (RMSE 6.49 mg/dL, MARD 3.10%, 100% Zone A), but this

is measured on a narrow (~ 40 mg/dL) single-subject range, which is well known to produce optimistic-looking MARD/RMSE relative to a wide-range, multi-subject clinical evaluation; the numbers in Table 2 should therefore be read as an in-sample, single-subject, proof-of-concept indicator and not compared directly to ISO 15197 CGM benchmarks.

Second, the real-time firmware equation (Fig. 4) performs materially worse than the offline Random Forest model: it tracks the approximate glycaemic level but not its fluctuations. This gap between an offline, feature-rich ensemble model and a firmware-simple single-ratio linear equation is expected and is exactly why the Random Forest results are presented as a research-stage comparison rather than as the currently deployed estimator (Section 2.4).

Third, bench evidence shows the current finger-presence gate does not reject optical saturation in real time (IR pinned at the 262,143-count ADC ceiling was observed being converted into a glucose value by the live firmware). This is a concrete firmware defect, not a theoretical concern, and should be fixed before any further human-subject data collection: the gate should reject samples with IR above a high threshold (e.g., $IR > \sim 255,000$ counts) in addition to the existing low-IR rejection.

Consistent with current exploratory approaches by major wearable manufacturers, this system is explicitly classified as a non-invasive, trend-monitoring proof-of-concept and not a diagnostic substitute for capillary blood glucose testing. Key limitations: (i) a largely single-subject local dataset supplemented by domain-shifted public data, limiting generalizability; (ii) a narrow tested glycaemic range not yet validated against hypoglycaemic or markedly hyperglycaemic conditions; (iii) no physiological compensation term for skin temperature or perfusion pressure yet; and (iv) the real-time firmware equation and the offline Random Forest model currently perform very differently, and this gap must be closed (or clearly scoped) before any claim of a working real-time non-invasive glucose product.

Future work: (1) fix the real-time saturation gate described above; (2) run a statistically powered end-to-end telemetry latency benchmark for the n8n/Supabase pipeline; (3) expand the calibration dataset across more subjects and a wider glycaemic range; (4) add temperature- and pressure-based compensation terms to the optical ratio, following the approach validated for multisensor non-invasive CGM by Zanon et al. (2013); and (5) evaluate the firmware equation and the Random Forest model on the same held-out data so that a single, defensible accuracy claim can be made for whichever model is ultimately deployed.

5. Conclusion

This work demonstrates the feasibility of repurposing a MAX30102/MAX30105 pulse-oximetry sensor as a dual-wavelength optical probe for non-invasive glucose-trend estimation within the Life Guard wearable ecosystem, combining ESP32-side signal conditioning, offline saturation rejection, and a data-fusion calibration strategy. An offline Random Forest model reaches strong single-subject agreement with reference glucose (RMSE 6.49 mg/dL, MARD 3.10%, 100% Zone A), while the currently deployed real-time firmware equation is shown, transparently, to under-perform this offline result and to require the saturation-gate fix identified in Section 4. The system is positioned as an early-stage, non-diagnostic proof-of-concept and a practical foundation for larger-scale validation, not as a finished continuous glucose monitor.

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